

Series Resistance — Debug Notes

Investigation into why the Series Resistance test returns $\sim 2.0\text{--}2.1\ \Omega$ on every pin / every matrix configuration, with variation only in the 2nd–3rd digit.

Files involved:

- `core/series_resistance_engine.py` — `SeriesResistanceRunner._run_row`
 - `core/engine.py` — base `SweepRunner` (`_apply_voltage`, `_measure`, `_setup_smu`)
 - `settings/series_resistance.json` — SMU config
 - `keithley4200.py` — driver (`apply()` dispatch on `self.source`)
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TL;DR

The measurement chain looks healthy. The matrix is switching, the cathode is force-sourcing current, and the measured voltage tracks the current change. The constant $\sim 2\ \Omega$ is the genuine output of the formula, not an artifact of a broken matrix or an open circuit. The two things that *looked* like bugs are not:

1. `_apply_voltage` being used for a "Current in A" channel — **not a bug**. The driver dispatches on sweep mode, so it force-sources current.
2. The + sign in the R_s formula — **not a bug**. It is correct *precisely* *because* the sweep currents (and the resulting voltages) are negative.

The real caveats are about **interpretation and robustness**, not correctness (see "Open risks" below).

1. `_apply_voltage` is a misnomer, not a dispatch bug

`series_resistance_engine.py` calls `self._apply_voltage(primary, step.voltage)` for the current-sourced cathode. That is fine:

```
# core/engine.py
def _apply_voltage(self, ch, voltage):
    smu = self.smu_drivers[ch.role]
    smu.value = voltage
    smu.apply()

# keithley4200.py
def apply(self):
    if self.source == "Voltage in V":
        self.set_voltage(...) # DV
    elif self.source == "Current in A":
        self.set_current(...) # DI <-- taken for cathode_SMU1
```

`self.source` comes from `SweepMode` → `ch.sweep_mode` → JSON "sweep_mode". For `cathode_SMU1` that is "Current in A", so `apply()` emits a DI command and **forces current**. `SweepStep.voltage` is just a generically-named numeric field; here it carries amps (-0.02 , -0.035). Confirmed by the live output:

Applied Cathode SMU1 = -0.020 A
Measured Cathode SMU1: -0.611389 V, -0.0199978 A

The forced current comes back as the measured current ($-0.0199978 \approx -0.020$ A), and a real junction voltage develops (-0.611 V). Everything is working.

Naming trap only: `_apply_voltage`, `SweepStep.voltage`, and the "hold at ... A" vs "... V" prints are misleading but functionally correct.

2. Difference vs Dark Current (which works)

	Dark Current	Series Resistance
Primary sweep_mode	absent $\rightarrow$ "Voltage in V"	"Current in A"
Primary action	force V, measure I	force I, measure V
Primary compliance_A	0.15 (current limit)	4.0 $\rightarrow$ voltage compliance under DI
Primary range	"Limited 1 nA" (RG)	"Auto" (no range cmd; DI range arg = 0)

When current-sourcing, the JSON `compliance_A: 4.0` is passed as the 4th DI argument = **voltage compliance (4 V)**, not a current limit. Measured V is ~ 0.6 V, well under 4 V, so we are *not* railing into compliance. Good.

3. The formula, with the real numbers

$$rs = ((v2 - v1) + 2.0 * \text{math.log}(i2 / i1) * 26e-3) / (i2 - i1)$$

This is the diode model $V = n \cdot V_t \cdot \ln(I/I_s) + I \cdot R_s$ with $n = 2$, $V_t = 26$ mV, solved for R_s across two operating points.

Row 1 numbers:

$$\begin{aligned} v1 &= -0.611389 \text{ V} & i1 &= -0.0199978 \text{ A} \\ v2 &= -0.671042 \text{ V} & i2 &= -0.0349996 \text{ A} \end{aligned}$$

$$\begin{aligned} \Delta v &= v2 - v1 &= -0.059653 \text{ V} \\ \Delta i &= i2 - i1 &= -0.0150018 \text{ A} \\ \ln(i2/i1) &= \ln(1.75018) &= 0.559725 \\ \text{thermal} &= 2 \cdot 0.026 \cdot \ln &= 0.029106 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{num} &= \Delta v + \text{thermal} = -0.059653 + 0.029106 = -0.030547 \\ rs &= \text{num} / \Delta i = -0.030547 / -0.0150018 = 2.036 \text{ } \Omega \quad \checkmark \text{ (matches printed 2.0363)} \end{aligned}$$

Why the + is correct (your I1/I2 question)

The textbook form *subtracts* the thermal term:

$$Rs = [(V2-V1) - n \cdot V_t \cdot \ln(I2/I1)] / (I2 - I1)$$

That assumes forward bias with **positive** I and V. Here the cathode sources **negative** current, so V is negative too. Define magnitudes $V' = -V$, $I' = -I$ (both positive). Then:

$$\begin{aligned} \ln(i2/i1) &= \ln(I2'/I1') && \text{(the minus signs cancel)} \\ (v2 - v1) &= -(V2' - V1') \end{aligned}$$

$$(i_2 - i_1) = -(I_2' - I_1')$$

```
code = [ (v2-v1) + n·Vt·ln(i2/i1) ] / (i2-i1)
       = [ -(V2'-V1') + n·Vt·ln(I2'/I1') ] / [ -(I2'-I1') ]
       = [ (V2'-V1') - n·Vt·ln(I2'/I1') ] / [ (I2'-I1') ] ← textbook form
```

So the + in the code is the - of the standard formula, after the two negatives in numerator and denominator cancel. **The negative sweep currents are exactly why the formula reads +.**

⚠ This makes the formula **convention-locked**: it is only correct while I_1, I_2 are negative. If anyone changes the sweep to positive currents, the + becomes wrong and must flip to -. (Sanity check: forcing - here would yield $5.92\ \Omega$, which is the *incorrect* value for this negative-current setup.)

4. Why it's the same $\sim 2\ \Omega$ on every pin (not necessarily a bug)

Two reasons, both expected:

1. **The DUTs are near-identical diodes.** Their forward I–V curves are highly reproducible, so V_1, V_2 barely change pin-to-pin. The matrix *is* distinguishing pins — Row 1 = $2.0363\ \Omega$, Row 2 = $2.0402\ \Omega$ — the 2nd/3rd-digit variation is real device-to-device spread, not noise floor.
 2. **The result is dominated by the model constants, not the ohmic drop.**
 R_s is the *small difference* of two comparable quantities:
 $\Delta v = -0.0597\ \text{V vs thermal} = -0.0291\ \text{V}$. The numerator (-0.0305) is heavily shaped by the hardcoded $n = 2$ and $V_t = 26\ \text{mV}$. Change n to 1 and the same data gives $\sim 3.0\ \Omega$ instead of $2.0\ \Omega$. So the measurement is only weakly sensitive to the device and strongly sensitive to the assumed diode model — which is *why* you see a near-constant number.
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5. Cross-validation: force-V (Dark Current) reproduces the force-I points

To check that the current source, matrix path, and voltage measurement are all self-consistent, we ran `settings/dark_current_series.json` — same pins, same matrix configs — but **voltage-driven** at the voltages R_s measured ($-0.611\ \text{V}, -0.670\ \text{V}$), and measured the resulting current.

```
python sweep.py .\settings\dark_current_series.json --limit-rows 2
```

5a. Reciprocity holds (cathode)

Forced V	Measured I (DC test)	Force-I reference (R_s test)
$-0.611\ \text{V}$	$-0.0198868\ \text{A}$ (Row 1) / $-0.0197798\ \text{A}$ (Row 2)	$-0.0199978\ \text{A}$ @ $-0.611389\ \text{V}$
$-0.670\ \text{V}$	$-0.0346819\ \text{A}$ (Row 1) / $-0.0345338\ \text{A}$ (Row 2)	$-0.0349996\ \text{A}$ @ $-0.671042\ \text{V}$

Force-V and force-I land on the **same I–V point**, confirming the chain is self-consistent. The small $\sim 0.5\ \%$ shortfall ($-19.9\ \text{mA}$ vs $-20.0\ \text{mA}$) is fully explained by the voltage difference: the DC run was programmed at $-0.611\ \text{V}$ but

the R_s run actually sat at -0.611389 V ($\approx 0.36\text{ mV}$ more forward bias). With the diode slope $dI/dV = I/(n \cdot V_t) \approx 0.02/0.052 \approx 0.385\text{ A/V}$, that 0.36 mV predicts $\approx 0.14\text{ mA}$ less current $\rightarrow \approx -19.86\text{ mA}$, matching the measured -19.89 mA . The methods agree to within the programmed-voltage rounding. ✓

5b. NEW finding: the "fixed" 0 V electrodes carry milliamps (return path)

The Dark Current run measures **all** channels (the R_s engine only measures the primary), which exposes something the R_s test hid. Anode/Guard/Group are held at 0 V yet each sources several mA, and they **sum to the cathode current (KCL)**:

Row 1, step 0 (cathode = -19.887 mA):

Anode $2.40927 + \text{Guard } 5.96859 + \text{Group } 11.5131 = 19.891\text{ mA} \approx |\text{cathode}| 19.887\text{ mA}$ ✓

Row 1, step 1 (cathode = -34.682 mA):

Anode $3.77478 + \text{Guard } 12.4065 + \text{Group } 18.5056 = 34.687\text{ mA} \approx |\text{cathode}| 34.682\text{ mA}$ ✓

Row 2 sums match too (19.784 vs 19.780 mA ; 34.539 vs 34.534 mA), and the split differs by pin (e.g. Anode 2.41 mA on OUT1 vs 1.36 mA on IN2), confirming the matrix really does route differently per pin.

Implication for R_s . The cathode current does not return through a single 2-terminal path — it splits across three grounded electrodes (anode || guard || group). So the voltage the R_s test measures at the cathode is across the composite cathode $\rightarrow$ (anode || guard || group) network, and the extracted $\sim 2\ \Omega$ is the series resistance of *that whole network*, not a clean two-terminal device R_s . This is almost certainly the bigger reason the value looks "the same everywhere": it's dominated by shared, pin-independent return paths plus the model constants from §4, not by the individual DUT.

Note: range: "Limited 1 nA" on the cathode/anode did not clamp these mA-level currents — limited range only sets the autorange floor, and compliance is 0.15 A , so autoranging up to tens of mA works fine. (It is slow, though.)

Open risks / recommendations (no code changed yet)

- **Convention-locked sign.** Document, or make robust, the assumption that $I_1, I_2 < 0$. A future "make currents positive" change silently breaks R_s .
- **Hardcoded $n = 2, V_t = 26\text{ mV}$.** No temperature compensation, and n directly sets the answer. Consider making them explicit config, and/or deriving V_t from temperature.
- **Two-point extraction.** With only two current points you cannot validate linearity or separate R_s from ideality. A multi-point current sweep + linear fit of dV/dI (or a proper R_s -extraction at higher currents where $I \cdot R_s$ dominates the log term) would be far more trustworthy.
- **Sensitivity.** Because R_s is a difference of two similar terms divided by a small Δi , a $\sim 1\%$ error in ΔV or in the assumed $n \cdot V_t$ moves R_s by a few percent. The current $20/35\text{ mA}$ spacing keeps the log term comparable to the

ohmic term; pushing to higher currents (or wider spacing) improves the SNR of the ohmic part.

- **Multi-terminal return path (from §5b).** The cathode current returns through anode || guard || group, not a single electrode. The Rs engine only measures the primary, so it cannot see this. If a true 2-terminal device Rs is wanted, the measurement topology (which electrode is the return, what the guard does) needs to be defined deliberately — otherwise $\sim 2\ \Omega$ reflects the shared network.

Verdict

No bug found in the SMU dispatch, the matrix path, or the formula sign. The force-V cross-check (§5) confirms the current source, matrix routing, and measurement are all self-consistent and reciprocal. The $\sim 2\ \Omega$ is a real result, but it is dominated by (a) the assumed diode model constants (n , v_t , §4) and (b) a multi-electrode return path (§5b) — not by per-device series resistance. If a true per-device Rs is the goal, the fix is in **methodology and topology** (multi-point fit, explicit/temperature-aware diode constants, a defined two-terminal return path), not in correcting a code defect.